

The net input to a cluster unit Y_i is computed as

$$y_in_i = y_out_i - \delta \sum_{j \neq i} y_out_j \quad (9.2)$$

As the clustering process advances, more and more clustering units get deactivated (by attaining an activation of value 0) until all units except one are deactivated. The only one remaining positively activated unit is the winner. The procedure followed by MAXNET to identify the cluster to which a given pattern $\mathbf{x} = [x_1, \dots, x_m]$ belongs is described in **Algorithm** MAXNET-Clustering (Fig. 9.2). Fig. 9.3 presents a flowchart of the procedure. Example 9.1 illustrates the procedure followed by a MAXNET.

Example 9.1 (Clustering by MAXNET)

Let us consider the 4-unit MAXNET shown in Fig. 9.4. The inhibitory weight is taken as $\delta = 0.2$ which satisfies the condition $0 < \delta < \frac{1}{4}$. The input pattern $\mathbf{x} = [x_1, x_2, x_3, x_4] = [0.5, 0.8, 0.3, 0.6]$ is to be clustered. The step-by-step execution of **Algorithm** MAXNET-Clustering is given below.

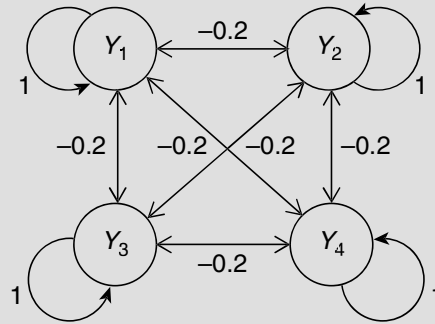


Fig. 9.4. A 4-unit MAXNET

Step 0. Initialize the net inputs to the cluster units.

$$y_in_1 = 0.5, y_in_2 = 0.8, y_in_3 = 0.3, y_in_4 = 0.6$$

Iteration #1

Step 1. Compute the activation of each cluster unit.

$$y_out_1 = 0.5, y_out_2 = 0.8, y_out_3 = 0.3, y_out_4 = 0.6.$$

Step 2. Test for stopping condition. **If** all units except one have 0 activation **Then** return the unit with non-zero activation as the winner. **Stop.**

There are four units with non-zero activations. Therefore the stopping criterion (that all units except one have zero activations) is not satisfied. Hence, continue to Step 3.

Step 3. Update the net input to each cluster unit.

$$y_in_1 = y_out_1 - \delta \sum_{j \neq 1} y_out_j = 0.5 - 0.2 \times (0.8 + 0.3 + 0.6)$$

$$= 0.5 - 0.2 \times 1.7 = 0.5 - 0.34 = 0.16$$

Similarly,

$$y_{in_2} = 0.8 - 0.2 \times (0.5 + 0.3 + 0.6) = 0.8 - 0.28 = 0.52$$

$$y_{in_3} = 0.3 - 0.2 \times (0.5 + 0.8 + 0.6) = 0.3 - 0.38 = -0.08$$

$$y_{in_4} = 0.6 - 0.2 \times (0.5 + 0.8 + 0.3) = 0.6 - 0.32 = 0.28$$

Step 4. Go to Step 1.

Iteration #2

Step 1. Compute the activation of each cluster unit.

$$y_{out_1} = 0.16, y_{out_2} = 0.52, y_{out_3} = 0, y_{out_4} = 0.28$$

Step 2. Test for stopping condition. **If** all units except one have 0 activation **Then** return the unit with non-zero activation as the winner. **Stop.**

There are three units with non-zero activations. Therefore the stopping criterion (that all units except one have zero activations) is not satisfied. Hence, continue to Step 3.

Step 3. Update the net input to each cluster unit.

$$\begin{aligned} y_{in_1} &= y_{out_1} - \delta \sum_{j \neq 1} y_{out_j} = 0.16 - 0.2 \times (0.52 + 0 + 0.28) \\ &= 0.16 - 0.2 \times 0.8 = 0.16 - 0.16 = 0 \end{aligned}$$

Similarly,

$$y_{in_2} = 0.432$$

$$y_{in_3} < 0$$

$$y_{in_4} = 0.144$$

Step 4. Go to Step 1.

The calculations till the net comes to a halt is shown in Table 9.1. We see that the net successfully identifies the unit Y_2 as having maximum input as all the other activations have been reduced to 0. Hence Y_2 is the winner and the given input pattern is clustered to the cluster unit Y_2 .

Table 9.1. Clustering by MAXNET

#	y_{in}				y_{out}			
	y_{in_1}	y_{in_2}	y_{in_3}	y_{in_4}	y_{out_1}	y_{out_2}	y_{out_3}	y_{out_4}
0	0.5	0.8	0.3	0.6	0.5	0.8	0.3	0.6
1	0.16	0.52	< 0	0.28	0.16	0.52	0	0.28
2	0	0.432	< 0	0.144	0	0.432	0	0.144
3	< 0	0.403	< 0	0.058	0	0.403	0	0.058
4	< 0	0.391	< 0	< 0	0	0.391	0	0